

Self-Consistent Earthquake Fault-Scaling Relations: Update and Extension to Stable Continental Strike-Slip Faults

by Mark Leonard

Abstract Using seismic moment (M_0)–length (L) data for stable continental region (SCR) faults, augmented by data not included in previous studies, this article reassesses and confirms the bilinear scaling relation for the dip-slip faults of Leonard (2010). Furthermore, using the new data, I propose a separate bilinear relation for estimating magnitude from surface rupture length. There are now a small number of SCR strike-slip earthquakes for which the fault dimensions have been estimated. I use these to constrain a trilinear scaling relation for this class of fault.

For the four fault types (interplate dip slip, interplate strike slip, SCR/intraplate dip slip, and SCR/intraplate strike slip), the scaling relations for the 10 permutations of M_0 , D , L , W , and A are given.

Introduction

In recent years, the fault rupture dimensions for several stable continental region (SCR) earthquakes have been estimated. As previous scaling relations in SCR (e.g., Johnston, 1994; Leonard, 2010) used fewer than 10 earthquakes, even a few extra earthquakes have the potential to better constrain the relations and enable new relations to be developed.

Strike-slip earthquakes do occur in the SCR, with studied examples in Portugal (Rockwell *et al.*, 2009) and Brazil (Assumpção *et al.*, 2014). The lack of data has prevented the development of scaling relations for SCR strike-slip faults, forcing other scaling relations to be applied, such as SCR dip slip (Wheeler, 2014). In a recent review, Stirling *et al.* (2013) discussed the importance of using tectonic and fault-style-specific scaling relations. I use data for nine SCR strike-slip earthquakes, six of which are new, to constrain scaling parameters for this class of fault.

To describe the relationship between fault width and length, Leonard (2010) proposed the relation

$$W = C_1 L^\beta, \quad (1)$$

in which W is the fault width and L is the fault length that applies to all fault types of length greater than around 2 km. For dip-slip faults, it applies up to the largest faults; and, for strike-slip faults, it applies up to lengths of 50–100 km, in which the faults become width limited. He demonstrated that, to a first-order approximation, $\beta = 2/3$ (though values between 0.59 and 0.725 provided reasonable fits to the data) and that C_1 varies between 13.5 and 17.5, depending on fault type. Leonard (2010) also demonstrated that the relation

$$D = C_2 \sqrt{A}, \quad (2)$$

(in which D is the average displacement and A is the fault area) provided a good fit to the data, though displacement functions between $D = C_2 W$ and $D = C_2 L$ also provided reasonable fits.

Leonard (2010) incorporated these two relations into the definition of seismic moment,

$$M_0 = \mu DA = \mu DLW, \quad (3)$$

in which $A = LW$ and μ is the average shear modulus (taken here as $3.3 \times 10^{10} \text{ N}\cdot\text{m}^{-2}$).

Doing so enabled them to develop a series of self-consistent scaling relations that depend only on β , C_1 , and C_2 . These relations enable any one of M_0 , D , L , W , and A to be determined from any of the others. Values of C_1 and C_2 were estimated from empirical data for interplate dip slip, interplate strike slip, and SCR dip-slip faults. Leonard (2010, 2012) gives the results for 7 of the 10 permutations of M_0 , D , L , W , and A for these three fault types. In this short note, the relations are given for all 10 permutations.

Data

Leonard (2010) relied on published datasets from Johnston (1994), Wells and Coppersmith (1994), Henry and Das (2001), Shaw and Scholz (2001), Hanks and Bakun (2002, 2008), Romanowicz and Ruff (2002), and Manighetti *et al.* (2007). These datasets contained only 10 SCR dip-slip earthquakes and 3 SCR strike-slip faults. The recent new data expand this to 16 SCR dip-slip earthquakes and 10 SCR strike-slip faults, which are summarized in Table 1. The details for each earthquake and fault are given in the Appendix.

Table 1
The Fault Dimensions and Fault-Surface-Rupture Length for Stable Continental Region (SCR) Earthquakes

Location	Date (yyyy/mm/dd)	Event	Type	M_w	M_0 ($N \cdot m \times 10^{16}$)	M_0 Range ($N \cdot m \times 10^{16}$)	
						Minimum	Maximum
Extended continental crust	1819/06/16	Allah Bund (Kutch), India	TH	7.79	60,000	18,000	200,000
	1939/07/22	Ghana, Africa	SS	6.45	590	209	16630
	2001/01/26	Bhuj, India	TH	7.63	35,000	25,000	49000
Nonextended continental crust	1966/10/09	Sudan, Africa	SS	5.64	36	8.1	144
	1967/12/10	Koyna India	SS	6.27	320	180	460
	1968/10/14	Meckering, Australia	TH	6.54	820	410	1200
	1970/03/10	Calingiri, Australia	TH	5.71	46.2	3.5	616
	1979/06/02	Cadoux, Australia	TH	6.09	173	122	244
	1983/12/22	Guinea, Africa	N/SS	6.29	339	240	479
	1983/10/07	Goodnow, United States	TH	4.8	1.9	1.35	2.68
	1986/03/30	Marayat Creek, Australia	TH	5.71	45.7	32.4	64.6
	1988/01/22	Tennant Creek 1, Australia	TH	6.23	273	193	386
	1988/01/22	Tennant Creek 2, Australia	TH	6.30	346	245	489
	1988/01/22	Tennant Creek 3, Australia	TH	6.58	916	648	1290
	1988/01/22	Tennant Creek, Australia (Σ)*	TH	6.72	1535	1100	2100
	1989/12/25	Ungava, Canada	TH	5.97	112	79.3	158
	1993/09/29	Killari Latur, India	TH	6.17	221	156	312
	1994/08/06	Ellalong, Australia	TH	4.71	1.46	1.04	3.5
	1994/08/21	Eugowra, Australia	TH	3.9	0.09	0.065	0.18
	2002/04/20	Au Sable Forks, New York	TH	4.95	3.40	2.58	6.1
	2004/08/07	Siheung, Korea	SS	2.62	0.00107	0.00097	0.0015
	2007/10/09	Katanning, Western Australia	TH	4.71	1.46	1.32	2.07
	2008/04/18	Mt. Carmel, Illinois	SS	5.23	8.8	7.0	17
	2010/02/09	Siheung, Korea	SS	3.14	0.0065	0.0046	0.009
	2011/11/05	Oklahoma 1, United States	SS	5.0	4	2.8	5.6
	2011/11/06	Oklahoma 2, United States	SS	5.7	45	32	63
	2011/11/08	Oklahoma 3, United States	SS	5.0	4	2.8	5.6
	2011/08/23	Mineral, Virginia	TH	5.74	52.3	38	60.0
	2012/03/23	Ernabella, Australia	TH	5.33	12.5	10.9	15.5

*The three Tennant Creek earthquakes summed into a single event.

Method

By substituting equations (1)–(3) into each other, 11 relations can be readily derived. These, along with the solution for $\beta = 2/3$, are given in equations (4)–(14). The definition of stress drop used by Kanamori and Anderson (1975) is shown as equation (15), in which L_C is the characteristic rupture dimension. Assuming $L_C = \sqrt{A}$ and (for a simple circular fault; Kanamori and Anderson, 1975) $C = 2.42$ gives equation (16).

$$W = C_1 L^\beta = C_1 L^{2/3}, \quad (4)$$

$$W = C_1^{1/(1+\beta)} A^{\beta/(1+\beta)} = C_1^{3/5} A^{2/5}, \quad (5)$$

$$A = C_1 L^{1+\beta} = C_1 L^{5/3}, \quad (6)$$

$$D = C_2 A^{1/2}, \quad (7)$$

$$\log M_0 = \frac{3}{2} \log A + \log C_2 \mu, \quad (8)$$

$$\begin{aligned} \log M_0 &= \frac{3}{2} (1 + \beta) \log L + \frac{3}{2} \log C_1 + \log C_2 \mu \\ &= \frac{5}{2} \log L + \frac{3}{2} \log C_1 + \log C_2 \mu, \end{aligned} \quad (9)$$

$$\begin{aligned} \log M_0 &= \frac{3}{2} \left(1 + \frac{1}{\beta} \right) \log W - \frac{3}{2\beta} \log C_1 + \log C_2 \mu \\ &= \frac{15}{4} \log W - \frac{9}{4} \log C_1 + \log C_2 \mu, \end{aligned} \quad (10)$$

$$\log M_0 = 3 \log \bar{D} - 2 \log C_2 + \log \mu, \quad (11)$$

$$\log \bar{D} = \frac{1}{2} \log A + \log C_2, \quad (12)$$

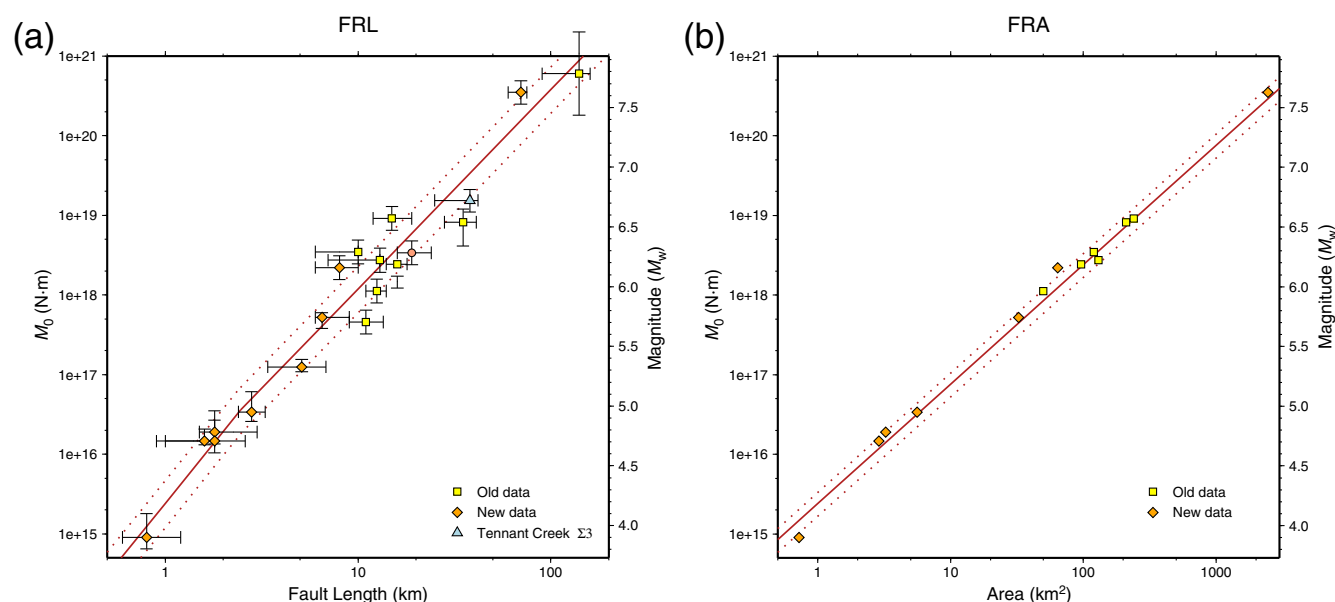


Figure 1. Scaling relations for stable continental region (SCR) dip-slip earthquakes: (a) magnitude length and (b) magnitude area. Note the slope of the regression lines was fixed by the choice of β and slip model (equation 2), both of which are set from interplate not SCR data. The regressions shown are from Leonard (2010) and were based only on the old data. The color version of this figure is available only in the electronic edition.

$$\begin{aligned}\log \bar{D} &= \frac{(1+\beta)}{2} \log L + \frac{1}{2} \log C_1 + \log C_2 \\ &= \frac{5}{6} \log L + \frac{1}{2} \log C_1 + \log C_2,\end{aligned}\quad (13)$$

$$\begin{aligned}\log \bar{D} &= \frac{1+1/\beta}{2} \log W + \frac{(-1/\beta)}{2} \log C_1 + \log C_2 \\ &= \frac{5}{4} \log W - \frac{3}{4} \log C_1 + \log C_2,\end{aligned}\quad (14)$$

$$\Delta\sigma = C\mu\left(\frac{\bar{D}}{L_C}\right) = \frac{CM_0}{AL_C},\quad (15)$$

and

$$\Delta\sigma \approx 8 \times 10^{10} C_2 \quad \text{when } W = L. \quad (16)$$

To estimate β and C_1 , Leonard (2010) applied a least-squares analysis to the interplate dip-slip $W \sim L$ empirical fault data using equation (4). β was consistent with the data, so it was fixed and a constrained least-squares analysis was used to estimate C_1 for the interplate strike-slip and SCR dip-slip faults. To estimate C_2 , Leonard (2010) used the $M_0 \sim A$ and $M_0 \sim L$ data (equations 8 and 9). In equations (8) and (9), the slope is independent of C_1 and C_2 ; therefore, constrained least-squares analysis was used to estimate C_2 . Leonard (2010) developed a bilinear relation, between $\log(L)$ and $\log(M_0)$, for dip-slip faults and a trilinear relation for strike-slip faults.

Because $\beta = 2/3$ (0.59–0.725) was found to apply to interplate dip slip, interplate strike-slip, and SCR dip slip, I assume it also applies to SCR strike-slip faults. This allows

a constrained least-squares analysis to be used to estimate C_1 and C_2 from the relatively small number of $\log(M_0)$ – $\log(L)$ data points. As the two small Korean earthquakes fall in the first section of the trilinear $\log(M_0)$ – $\log(L)$ relation, C_2 can be estimated independently of C_1 . (In the first section, $W = L$, so $\beta = 1.0$ and $C_1 = 1.0$; therefore, equation 1 becomes independent of C_1 .)

Strike-slip earthquakes become width limited once the width of the fault reaches the seismogenic depth and so scale differently (Scholz *et al.*, 1986; Shaw and Scholz, 2001; Romanowicz and Ruff, 2002). For the Australian continent, Leonard (2008) showed the vast majority of well-resolved earthquakes were at a depth of < 20 km, with a few possibly extending to 25 km. Bent and Perry (2002) and Mazzotti and Adams (2005) found similar distributions for eastern Canada. There are a small number of nonextended SCR earthquakes deeper than 20 km (e.g., Ansell *et al.*, 1986). Klose and Seeber (2007) proposed a bimodal model of depth distribution with most earthquakes in the upper third (0–10 km) and/or lower third (10–30 km), with some earthquakes (e.g., 2001 M_w 7.6 Bhuj) rupturing the entire crust. Mazzotti and Adams (2005) used three evenly weighted depths of 12, 21, and 30 km in their analysis of deformation rates in eastern Canada. I chose 20 km as an appropriate average maximum seismogenic depth in SCR. Below, I discuss options in which the seismogenic depth is more or less than 20 km.

Results

The updated SCR dip-slip $\log(M_0)$ – $\log(L)$ and $\log(M_0)$ – $\log(A)$ data (Fig. 1) is fully consistent with the original bilinear relation of Leonard (2010), with both the

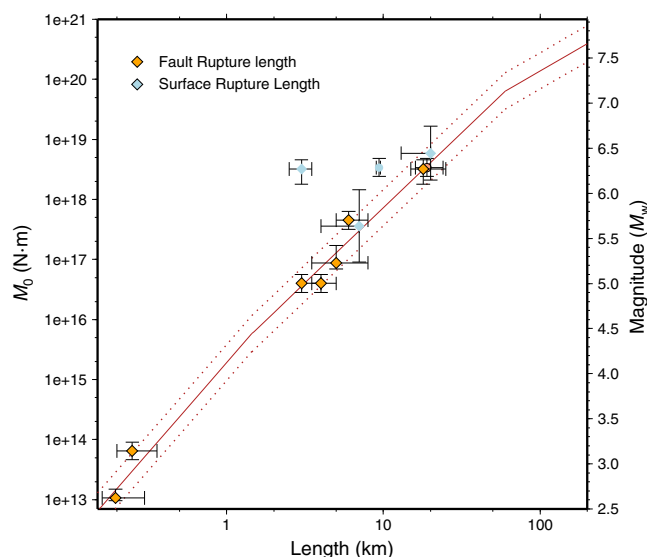


Figure 2. The trilinear magnitude–length relation for SCR strike-slip earthquakes. The color version of this figure is available only in the electronic edition.

best-fit relation and its uncertainties being essentially identical. As such, the predictive equations of Leonard (2010) remain valid.

The fitting of the trilinear SCR strike-slip relation to the $\log(M_0)$ – $\log(L)$ data is shown in Figure 2. The parameters C_1 , C_2 , and the corner length (~ 1.2 km) are reasonably well constrained from the data. The second corner at 70 km is constrained by the assumption of a maximum fault width of 20 km. Leonard (2010) showed the relation $D = C_2\sqrt{A}$ provides a good fit to the data beyond the second corner leading to the relation $\log M_0 = 1.5 \log L + C_3$. $C_3 = \log(\mu C_2 W^{3/2})$ has been set such that this relation meets $\log M_0 = 2.5 \log L + C_2$ at the L corresponding to a maximum width fault; in this case, $L = 70$ km, for which $M = 7.2$. It would be straightforward to modify these relations for longer or shorter values of L . I note that for a maximum seismogenic width of 30 and 40 km, $L = 130$ and 200 km, for which $M = 7.7$ and $M_w = 8.1$, respectively. As $M_{\max}$ in SCR is likely between $M_w 7.2$ and $M_w 7.7$ (Wheeler, 2009; Leonard and Clark, 2011; Leonard *et al.*, 2014), width-limited SCR strike-slip earthquakes will be rare. The results for the four fault types are presented in Table 2.

Table 2

The Estimated Parameters C_1 and C_2 , for $\beta = 2/3$, that are Used in Equations (4)–(16), for Each of the Four Fault Types

Fault Type	C_1 (m ^{1/3})	$C_2 \times 10^{-5}$
Interplate dip slip	17.5 (12–25)	3.8 (1.5–12)
Interplate strike slip	15.0 (11–20)	3.7 (1.5–9.0)
SCR dip slip	13.5 (10–17)	7.3 (5.0–10)
SCR strike slip	11.3 (9–15)	5.8 (3.0–11)

Scaling Relations for Surface Rupture Length

The scaling relations discussed so far exclusively involve the length of fault rupture (L_{FR}). Scaling relations involving surface rupture length (L_{SR}) have additional complications. From the definition of seismic moment and assuming a linear, or even power, relationship between displacement and a fault dimension (i.e., r , W , $\sqrt{A}$, L), a linear relationship is expected between $\log(M_0)$ and $\log(L)$ or A or W or D). The relationship between $\log(M_0)$ and $\log(L_{SR})$ is dependent on the relationship between L_{FR} and L_{SR} . L_{SR} is the length that can be measured at the surface, generally by a field party. It will depend on the ratio of fault width to hypocentral depth, the total displacement, the displacement profile (e.g., boxcar, triangular, $\cos^2$), the soil type, thickness and water content, the local topography, and the environment (e.g., rainfall, vegetation). I suggest these dependencies are so variable that it is not possible to develop a single simple relationship between L_{FR} and L_{SR} in all environments. For large, shallow earthquakes (e.g., $M_w = 6.5$, $L = 20$ km, $D = 1.0$ m), it is likely that $\geq 90\%$ of the fault length will be readily discoverable for a wide range of environments. Scarps caused by moderate earthquakes (e.g., $M_w = 5.5$, $L = 5$ km, $D = 0.3$ m) will only be discoverable under highly favorable circumstances and small earthquakes (e.g., $M_w = 4.5$, $L = 1.3$ km, $D = 0.09$ m) only under exceptional circumstances.

I use the L_{FR} – L_{SR} data for 10 SCR dip-slip faults to investigate the relationship between L_{FR} and L_{SR} in flat, arid to semiarid areas with thin soils, such as around Tennant Creek, Ernabella, and Katanning. For these data, I suggest $L_{SR} = 0.13 \times L_{FR}^{1.75}$ for $L_{FR} < 15$ km is a reasonable fit to the data (Fig. 3). Substituting this into equation (9) gives $M_0 = 6.9 \times 10^{16} \times L_{SR}^{1.43}$, which equates to $M_w = 5.16 + 0.953 \log(L_{SR})$. Figure 4 shows that this is a surprisingly good fit to the $\log(L_{SR})$ – $\log(M_0)$ data for $L_{FR} < 15$ km. Previous $\log(L_{SR})$ – M_w studies by Johnston (1994) and Clark *et al.* (2014) are intermediate between the two segments of our bilinear relationship.

Discussion

The various $\log M_0$, A , L , W , and D relations are given in Table 3 and the M_w relations are given, using $M_w = (\log(M_0) - 9.1)/1.5$ (Kanamori and Anderson, 1975; Kanamori, 1978; Purcaru and Berckhemer, 1978), in Table 4. There are slight variations between these values and the values given in Leonard (2010). As Leonard (2010) gave the regression values to only two decimal places, the multiple lines in the bilinear and trilinear curves did not exactly meet at the specified ranges. The regression values in Tables 3 and 4 are now given to three decimal places, and some of the ranges adjusted. The differences between the versions, for the average value, are < 0.01 magnitude units. The largest differences are in the uncertainty range of a in some relations

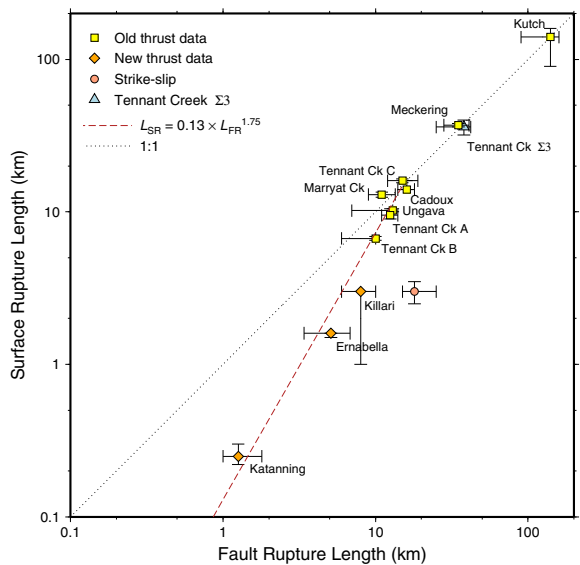


Figure 3. The SCR fault rupture length (L_{FR}) and surface rupture length (L_{SR}) data. I propose a bilinear relationship between L_{FR} and L_{SR} . For $L_{FR} \leq 15$ km, the dashed line is a reasonable fit to areas favorable to the creation of a fault scarp, such as the flat arid to semiarid environments at Katanning, Ernabella, and Tennant Creek. Above 15 km, $L_{FR} = L_{SR}$ fits the data. The color version of this figure is available only in the electronic edition.

(e.g., $\log D_{Av} = a + \log L$), which have changed by 1%–2%.

In the formulation of equations (4)–(14), as $\log X = a + b \log Y$, the b -value is dependent only on β and the displacement function (i.e., equation 2), not on C_1 or C_2 . Because $\beta = 2/3$ and equation (2) have been used in all tectonic environments and fault styles, the differences are controlled by the parameters C_1 and C_2 . Variations in parameter C_2 reflect variations in average stress drop between different

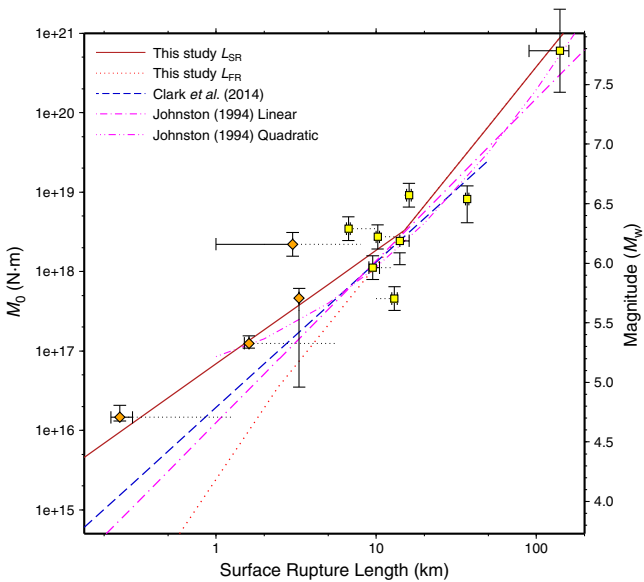


Figure 4. Comparison of the proposed $\log(L_{SR})$ – $\log(M_0)$ scaling relation with existing relations. The bilinear relation has not been fitted to this data but derived from the $\log(L_{FR})$ – $\log(L_{SR})$ relation shown in Figure 3. The relations of Johnston (1994) and Clark *et al.* (2014) are intermediate between this relation and the $\log(L_{FR})$ – $\log(M_0)$ relation of Leonard (2010). The dashed lines connect the L_{SR} – L_{FR} . The color version of this figure is available only in the electronic edition.

tectonic environments and fault styles. Variations in parameter C_1 reflect variations in the size of a fault when it transitions from $W = L$ to $W = C_1 L^\beta$.

Fault scaling relations fall into two classes: model driven (e.g., Hanks and Bakun, 2002; Romanowicz and Ruff, 2002; Shaw and Wesnousky, 2008; Leonard, 2010) and purely empirical (e.g., Wells and Coppersmith, 1994; Blaser *et al.*,

Table 3
Summary of the Various Moment-Based Fault-Scaling Relations

Fault Type	Relation	b	a	$S(a)^*$	Range $A(m^2)$, $L(m)$, $W(m)$, $D(m)^\dagger$
Interplate DS	$\log(A) = a + b \times \log(L)$	2.0	0.0	—	0–5360
		1.667	1.243	1.08 to 1.40	> 5360
	$\log(W) = a + b \times \log(L)$	1.0	0.0	—	0–5360
		0.667	1.243	1.08 to 1.40	> 5360
	$\log(W) = a + b \times \log(A)$	0.5	0.0	—	0–28.7 × 10 ⁶
		0.4	0.746	0.65 to 0.84	> 28.7 × 10 ⁶
	$\log(D_{Av}) = a + b \times \log(A)$	0.5	–4.420	–4.82 to –3.92	> 0
	$\log(D_{Av}) = a + b \times \log(L)$	1.0	–4.420	–4.90 to –3.82	0–5360
		0.833	–3.799	–4.28 to –3.22	> 5360
	$\log(D_{Av}) = a + b \times \log(W)$	1.25	–5.352	–5.63 to –4.97	> 5360
	$\log(M_0) = a + b \times \log(A)$	1.5	6.098	5.70 to 6.60	> 0
	$\log(M_0) = a + b \times \log(L)$	3.0	6.098	5.45 to 6.83	0–5360
		2.5	7.963	7.31 to 8.70	> 5360
	$\log(M_0) = a + b \times \log(W)$	3.75	3.301	3.27 to 3.45	> 5360

(continued)

Table 3 (Continued)

Fault Type	Relation	<i>b</i>	<i>a</i>	<i>S(a)*</i>	Range <i>A</i> (m ²), <i>L</i> (m), <i>W</i> (m), <i>D</i> (m) [†]
Interplate SS	$\log(A) = a + b \times \log(L)$	2.0	0.0	—	0–3400
		1.667	1.176	1.04 to 1.30	3400–40,000
		1.0	4.244	4.11 to 4.37	> 40,000
	$\log(W) = a + b \times \log(L)$	1.0	0.0	—	0–3400
		0.667	1.176	1.04 to 1.30	3400–40,000
		0.0	4.244	4.11 to 4.37	> 40,000
	$\log(W) = a + b \times \log(A)$	0.5	0.0	—	0–11.5 × 10 ⁶
		0.4	0.706	0.63 to 0.78	11.5–702 × 10 ⁶
		0.0	4.244	4.11 to 4.37	> 702 × 10 ⁶
	$\log(D_{Av}) = a + b \times \log(A)$	0.5	–4.432	–4.82 to –4.05	> 0
	$\log(D_{Av}) = a + b \times \log(L)$	1.0	–4.432	–4.82 to –4.05	0–3400
		0.833	–3.844	–4.30 to –3.40	3400–40,000
		0.5	–2.310	–2.77 to –1.86	> 40,000
	$\log(D_{Av}) = a + b \times \log(W)$	1.25	–5.314	–5.02 to –5.60	3400–17,500
	$\log(M_0) = a + b \times \log(A)$	1.5	6.087	5.69 to 6.47	> 0
	$\log(M_0) = a + b \times \log(L)$	3.0	6.087	5.49 to 6.66	0–3400
		2.5	7.851	7.41 to 8.28	3400–40,000
		1.5	12.45	11.86 to 13.03	> 40,000
	$\log(M_0) = a + b \times \log(W)$	3.75	3.441	3.35 to 3.54	3400–17,500
	$\log(M_0) = a + b \times \log(D)$	3.0	14.95	14.56 to 15.34	> 0
	$\log(M_0) = a + b \times \log(SRL)$	2.27	9.15		3400–65,000
		1.5	12.86		> 65,000
SCR DS	$\log(A) = a + b \times \log(L)$	2.0	0.0	—	0–2500
		1.667	1.130	1.04 to 1.23	> 2500
	$\log(W) = a + b \times \log(L)$	1.0	1.0	–	< 2500
		0.667	1.130	1.02 to 1.23	> 2500
	$\log(W) = a + b \times \log(A)$	0.5	0.0	—	0–6.2 × 10 ⁶
		0.4	0.678	0.63 to 0.74	> 6.2
	$\log(D_{Av}) = a + b \times \log(A)$	0.5	–4.137	–4.30 to –4.00	> 0
	$\log(D_{Av}) = a + b \times \log(L)$	1.0	–4.137	–4.30 to –4.00	0–2500
		0.833	–3.572	–3.78 to –3.38	> 2500
	$\log(D_{Av}) = a + b \times \log(W)$	1.25	–4.984	–4.92 to –5.08	> 2500
	$\log(M_0) = a + b \times \log(A)$	1.5	6.38	6.22 to 6.52	> 0
	$\log(M_0) = a + b \times \log(L)$	3.0	6.382	6.08 to 6.67	0–2500
		2.5	8.077	7.78 to 8.36	> 2500
	$\log(M_0) = a + b \times \log(W)$	3.75	3.84	3.75 to 3.87	> 2500
	$\log(M_0) = a + b \times \log(D)$	3.0	14.66	14.52 to 14.82	> 0
		1.43	12.55		0–15,000
SCR SS	$\log(A) = a + b \times \log(L)$	2.5	8.08		> 15,000
		2.0	0.0	–	0–1600
		1.667	1.068	0.996 to 1.16	1600–70,000
	$\log(W) = a + b \times \log(L)$	1.0	4.298	4.23 to 4.39	> 70,000
		1	0.0	—	0–1600
		0.667	1.068	0.996 to 1.16	1600–70,000
	$\log(W) = a + b \times \log(A)$	0.0	4.298	4.23 to 4.39	> 70,000
		0.5	0.0	–	0–2.56 × 10 ⁶
		0.4	0.641	0.60 to 0.70	2.56–1400
	$\log(D_{Av}) = a + b \times \log(A)$	0.0	4.298	4.23 to 4.39	> 1400
		0.5	–4.149	–4.01 to –4.31	> 0
		1.0	–4.149	–4.35 to –3.97	0–1600
	$\log(D_{Av}) = a + b \times \log(L)$	0.833	–3.615	–3.81 to –3.43	1600–60,000
		0.5	–2.022	–2.22 to –1.84	> 60,000
		1.25	–4.950	–4.89 to –5.06	1600–18,000
	$\log(M_0) = a + b \times \log(A)$	1.5	6.370	6.21 to 6.51	> 0
	$\log(M_0) = a + b \times \log(L)$	3.0	6.370	6.10 to 6.65	0–1600
		2.5	7.972	7.70 to 8.25	1600–60,000
		1.5	12.750	12.48 to 13.03	> 60,000
	$\log(M_0) = a + b \times \log(W)$	3.75	3.966	3.86 to 3.97	1600–18,000
	$\log(M_0) = a + b \times \log(D)$	3.0	14.67	14.53 to 14.83	> 0

DS, dip slip; SS, strike slip.

**S(a)* is the one standard deviation range of *a*.†The range applies to the range of *Y* in the formulation $\log(X) = a + b \times \log(Y)$.

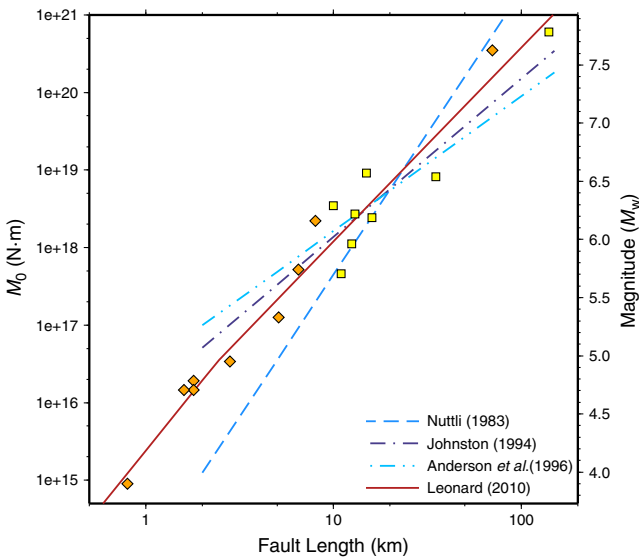


Figure 5. Comparison of published $\log(L_{FR})$ – $\log(M_0)$ relations for SCR dip-slip earthquakes with the augmented database. The color version of this figure is available only in the electronic edition.

2010; Clark *et al.*, 2014). Model-driven relations have the advantage that they are valid for the range of dimensions that the model is considered valid, whereas purely empirical models should only be used in the range of the data used to constrain them.

Most scaling relations give very similar results within the range of high-data density, for example, between magnitude 6.0 and 6.5 in SCR (see Fig. 5) and magnitudes 7.0–8.0

in subduction zones (compare Blaser *et al.*, 2010; Leonard, 2010; Strasser *et al.*, 2010). Given the variance for each of the four relations (in Fig. 5) is at least ± 0.25 , the differences between these relations at 20 km (M_w 6.5) are insignificant (± 0.05). Because the variability in earthquake scaling is of interest, I suggest taking a single scaling relation and formally including its uncertainty is preferable to taking the mean of multiple relations.

Although the proposed bilinear relation for surface-rupture faults provides a reasonable fit to the data, I estimate the uncertainties for $L_{SR} \leq 3.0$ km to be on the order of ± 0.5 magnitude units. Given these large uncertainties, I suggest that estimation of the magnitude of the causative earthquake from a mapped paleoscarp length of < 10 km, without any other geological or geophysical data (e.g., geophysical estimates of L_{FR} or average single event displacement) should be treated as at best indicative, even in a favorable environment.

Conclusion

Using newly available data, scaling relations for SCR strike-slip faults have been derived and previous SCR dip-slip scaling relations verified. For the four fault types (interplate dip slip and strike slip, and SCR/intraplate dip slip and strike slip), the scaling relations for the 10 permutations of M_0 , D , L , W , and A are given.

From an analysis of available fault rupture length (L_{FR}) and surface rupture length (L_{SR}) data, I propose a bilinear L_{FR} – L_{SR} relationship, which when substituted into equation

Table 4
Summary of the More Commonly Used Magnitude-Based Fault-Scaling Relations

Fault Type	Relation	<i>b</i>	<i>a</i>	<i>S</i> (<i>a</i>)	Range <i>A</i> (km ²), <i>L</i> (km), <i>W</i> (km), <i>D</i> (m)*
Interplate DS	$M_w = a + b \times \log(A)$	1.0	4.00	3.73–4.33	> 0
	$M_w = a + b \times \log(L)$	2.0	4.00		
		1.667	4.24	3.81–4.73	> 5.4
	$M_w = a + b \times \log(W)$	2.5	3.63	3.61–3.73	> 5.4
	$M_w = a + b \times \log(D)$	2.0	6.84	6.17–7.38	> 0
Interplate SS	$M_w = a + b \times \log(A)$	1.0	3.99	3.73–4.25	> 0
	$M_w = a + b \times \log(L)$	1.667	4.17	3.77–4.55	3.4–45.0
		1.0	5.27		> 45
	$M_w = a + b \times \log(W)$	2.5	3.88	3.82–3.95	3.4–19.0
	$M_w = a + b \times \log(D)$	2.0	6.85	6.34–7.38	> 0.13
SCR DS	$M_w = a + b \times \log(A)$	1.0	4.19	4.08–4.28	> 0
	$M_w = a + b \times \log(L)$	1.667	4.32	4.12–4.51	> 2.5
	$M_w = a + b \times \log(W)$	2.5	4.14	4.08–4.17	> 2.5
	$M_w = a + b \times \log(D)$	2.0	6.46	6.28–6.68	> 0.18
	$M_w = a + b \times \log(L_{SR})$	0.953	5.12		≤ 15
SCR SS		1.667	4.32		> 15
	$M_w = a + b \times \log(A)$	1.0	4.18	4.07–4.25	> 0
	$M_w = a + b \times \log(L)$	1.667	4.25	4.07–4.43	1.6–70
		1.0	5.44		> 60
	$M_w = a + b \times \log(W)$	2.5	4.22	4.17–4.23	1.6–20
	$M_w = a + b \times \log(D)$	2.0	3.71	3.62–3.82	> 0

DS, dip slip; SS, strike slip.
*Note the change of units, in the range, between Tables 3 and 4.

(9) gives a bilinear $\log(M_0)$ – $\log(L_{SR})$ relation, that is, a reasonable fit to the data. The uncertainty in this relation for $L_{SR} \leq 3.0$ km is greater than ± 0.5 magnitude units.

Data and Resources

All data used in this article are described in detail in the [Appendix](#). Generic Mapping Tool (Wessel and Smith, 1998) was used to generate the figures.

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References

- Abercrombie, R. E., and L. G. Baise (2005). Regional propagation in New England and New York: Collaborative research with Boston University and Tufts, *NEHRP Final Technical Report*, Award Number 04HQGR-0166.
- Adams, J., J. Percival, R. Wetmiller, J. Drysdale, and P. Robertson (1992). Geological controls on the 1989 Ungava surface rupture: A preliminary interpretation, *Geol. Surv. Can. Pap.* **92C**, 147–155.
- Adams, J., R. Wetmiller, H. Hasegawa, and J. Drysdale (1991). The first surface faulting from a historical earthquake in North America, *Nature* **352**, 617–619.
- Anderson, J. G., S. G. Wesnousky, and M. W. Stirling (1996). Earthquake size as a function of fault slip rate, *Bull. Seismol. Soc. Am.* **86**, no. 3, 683–690.
- Ansell, J., W. Aspinall, G. King, and R. Westaway (1986). The 1984 July 19 North Wales earthquake—A lower crustal continental event indicating brittle behaviour at an unusual depth, *Geophys. J. Int.* **84**, 201–206.
- Assumpção, M., J. Ferreira, L. Barros, H. Bezerra, G. S. Franca, J. R. Barbosa, E. Menezes, L. C. Ribotta, M. Pirchiner, A. D. Nascimento, and J. C. Dourado (2014). Intraplate seismicity in Brazil, in *Intraplate Earthquakes*, P. Talwani (Editor), Cambridge University Press, Cambridge, United Kingdom, 50–71.
- Bates, D. A. (1941). Geological effects of the [Accra] earthquake, *Gold Coast Geol. Surv. Bull.* **13**, 18–41.
- Bent, A. L. (1994). The 1989 (M_s 6.3) Ungava, Quebec, earthquake: A complex intraplate event, *Bull. Seismol. Soc. Am.* **84**, 1075–1088.
- Bent, A. L., and H. K. C. Perry (2002). Depths of eastern Canadian earthquakes from regional data, *Seismol. Res. Lett.* **73**, 273–284.
- Blaser, L., F. Krüger, M. Ohrnberger, and F. Scherbaum (2010). Scaling relations of earthquake source parameter estimates with special focus on subduction environment, *Bull. Seismol. Soc. Am.* **100**, no. 6, 2914–2926.
- Bowman, J. R. (1988). Constraints on locations of large intraplate earthquakes in the Northern Territory, Australia from observations at the Warramunga seismic array, *Geophys. Res. Lett.* **15**, 1475–1478.
- Bowman, J. R. (1991). Geodetic evidence for conjugate faulting during the 1988 Tennant Creek, Australia earthquake sequence, *Geophys. J. Int.* **107**, 47–56.
- Bowman, J. R. (1992). The 1988 Tennant Creek, Northern Territory, earthquakes: A synthesis, *Aust. J. Earth Sci.* **39**, 651–669.
- Bowman, J. R., G. Gibson, and T. Jones (1990). Aftershocks of the 1988 January 22 Tennant Creek, Australia intraplate earthquakes: Evidence for a complex thrust-fault geometry, *Geophys. J. Int.* **100**, 87–97.
- Brune, J. N. (1970). Tectonic stress and the spectra of seismic shear waves from earthquakes, *J. Geophys. Res.* **75**, 4997.
- Chapman, M. C. (2013). On the rupture process of the 23 August 2011 Virginia earthquake, *Bull. Seismol. Soc. Am.* **103**, 613–628.
- Choy, G. L., and J. R. Bowman (1990). Rupture process of a multiple main shock sequence; Analysis of teleseismic, local, and field observations of the Tennant Creek, Australia, earthquakes of January 22, 1988, *J. Geophys. Res.* **95**, 6867–6882.
- Clark, D., A. McPherson, T. Allen, and M. De Kool (2014). Coseismic surface deformation caused by the 23 March 2012 M_w 5.4 Ernabella (Pukatja) earthquake, central Australia: Implications for fault scaling relations in Cratonic settings, *Bull. Seismol. Soc. Am.* **104**, no. 1, 2368–2377.
- Copley, A., J. P. Avouac, J. Hollingsworth, and S. Leprince (2011). The 2001 M_w 7.6 Bhuj earthquake, low fault friction, and the crustal support of plate driving forces in India, *J. Geophys. Res.: Solid Earth* (1978–2012) **116**, B8, doi: [10.1029/2010JB008137](https://doi.org/10.1029/2010JB008137).
- Cramer, C. H., J. Kutliroff, and D. Dangkua (2011). The 2011 Mineral, VA M 5.8 earthquake ground motions and stress drop: An important contribution to the NGA east ground motion database, *American Geophysical Union, 2011 Fall Meeting*, San Francisco, California, 14–18 December 2009, Abstract S11B-2239.
- Crone, A. J., P. M. De Martini, M. N. Machette, K. Okumura, and J. R. Prescott (2003). Paleoseismicity of two historically quiescent faults in Australia: Implications for fault behavior in stable continental regions, *Bull. Seismol. Soc. Am.* **93**, 1913–1934.
- Crone, A. J., M. N. Machette, and J. R. Bowman (1992). Geologic investigations of the 1988 Tennant Creek, Australia, earthquakes—Implications for paleoseismicity in stable continental regions, *U.S. Geol. Surv. Bull.* **2032-A**, 51.
- Dawson, J., P. Cummins, P. Tregoning, and M. Leonard (2008). Shallow intraplate earthquakes in Western Australia observed by Interferometric Synthetic Aperture Radar, *J. Geophys. Res.* **113**, doi: [10.1029/2008JB005807](https://doi.org/10.1029/2008JB005807).
- Denham, D., L. G. Alexander, I. B. Everingham, P. J. Gregson, R. McCaffrey, and J. R. Enever (1987). The 1979 Cadoux earthquake and intraplate stress in Western Australia, *Aust. J. Earth Sci.* **34**, 507–521.
- Denham, D., L. G. Alexander, and G. Worotnicki (1980). The stress field near the sites of the Meckering (1968) and Calingiri (1970) earthquakes, Western Australia, *Tectonophysics* **67**, 283–317.
- Dorbath, C., L. Dorbath, R. Gaulon, T. George, P. Mourgue, M. Ramdani, B. Robineau, and B. Tadili (1984). Seismotectonics of the Guinean earthquake of December 22, 1983, *Geophys. Res. Lett.* **11**, 971–974.
- Ellsworth, W. L., K. Imanishi, J. H. Luetgert, J. Kruger, and J. Hamilton (2011). The M_w 5.8 Virginia earthquake of August 23, 2011 and its aftershocks: A shallow high stress drop event, *American Geophysical Union, 2011 Fall Meeting*, San Francisco, California, 14–18 December 2009, Abstract S14B-05.
- Gahalaut, V. K., Kalpna, and P. S. Raju (2003). Rupture mechanism of the 1993 Killari earthquake, India: Constraints from aftershocks and static stress change, *Tectonophysics* **369**, 71–78.
- Gibson, G., V. Wesson, and T. Jones (1994). The Eugowra NSW earthquake swarm of 1994, in *Proc. Australian Earthquake Engineering Society*, 14–15 November, Canberra, Australia.
- Gordon, F. R., and J. D. Lewis (1980). The Meckering and Calingiri earthquakes October 1968 and March 1970, *West. Aust. Geol. Surv. Bull.* **126**, 229 pp.
- Gupta, H. K., C. V. R. K. Rao, and B. K. Rastogi (1980). An investigation of earthquakes in Koyana region, Maharashtra for the period of October 1973 through December 1976, *Bull. Seismol. Soc. Am.* **70**, 1833–1847.
- Gupta, H. K., R. U. M. Rao, R. Srinivasan, G. V. Rao, G. K. Reddy, K. K. Dwivedy, D. C. Banerjee, R. Mohanty, and Y. R. Satyasradhi (1999). Anatomy of surface rupture zones of two stable continental region earthquakes, 1967 Koyana and 1993 Latur, India, *Geophys. Res. Lett.* **23**, 1985–1988.
- Hanks, T. C., and W. H. Bakun (2002). A bilinear source-scaling model for M – $\log A$ observations of continental earthquakes, *Bull. Seismol. Soc. Am.* **92**, no. 5, 1841–1846.
- Hanks, T. C., and W. H. Bakun (2008). M – $\log A$ observations for recent large earthquakes, *Bull. Seismol. Soc. Am.* **98**, no. 1, 490–494.
- Harsh, K., H. K. Gupta, B. K. Rastogi, I. Mohan, C. V. R. K. Rao, S. V. S. Sarma, and R. U. M. Rao (1998). An investigation into the Latur earthquake of September 29, 1993 in southern India, *Tectonophysics* **287**, 299–318.
- Hartzell, S., and C. Mendoza (2011). Source and site response study of the 2008 Mount Carmel, Illinois, earthquake, *Bull. Seismol. Soc. Am.* **101**, no. 3, 951–963.

- Haskell, N. A. (1966). Total energy and energy spectral density of elastic wave radiation from propagating faults. Part II. A statistical source model, *Bull. Seismol. Soc. Am.* **56**, no. 1, 125–140.
- Henry, C., and S. Das (2001). Aftershock zones of large shallow earthquakes: Fault dimensions, aftershock area expansion and scaling relations, *Geophys. J. Int.* **147**, 272–293.
- Herrmann, R. B., M. Withers, and H. Benz (2008). The April 18, 2008 Illinois earthquake: An ANSS monitoring success, *Seismol. Res. Lett.* **79**, 830–843.
- Johnston, A. C. (1994). Seismotectonic interpretations and conclusions from the stable continental region seismicity database, in *The Earthquakes of Stable Continental Regions—v. 1 Assessment of Large Earthquake Potential*, A. C. Johnston, K. J. Coppersmith, L. R. Kanter, and C. A. Cornell (Editors), Electric Power Research Institute Report TR102261V1, Palo Alto, California.
- Jones, T., V. Wesson, K. McCue, G. Gibson, C. Bricker, W. Peck, and A. Pascale (1994). The Ellalong, New South Wales, earthquake of 6 August 1994, *Proc. Australian Earthquake Engineering Society*, 14–15 November, Canberra, Australia.
- Kanamori, A. (1978). Quantification of great earthquakes, *Tectonophysics* **49**, 207–212.
- Kanamori, H., and D. L. Anderson (1975). Theoretical basis of some empirical relations in seismology, *Bull. Seismol. Soc. Am.* **65**, no. 5, 1073–1095.
- Keranen, K. M., H. M. Savage, G. A. Abers, and E. S. Cochran (2013). Potentially induced earthquakes in Oklahoma, USA: Links between wastewater injection and the 2011 M_w 5.7 earthquake sequence, *Geology* **41**, no. 6, 699–702.
- Kim, W.-Y., and K.-H. Kim (2014). The 9 February 2010 Siheung, Korea, earthquake sequence: Repeating earthquakes in a stable continental region, *Bull. Seismol. Soc. Am.* **104**, no. 1, doi: [10.1785/0120130119](https://doi.org/10.1785/0120130119).
- Klose, C. D., and L. Seeber (2007). Shallow seismicity in stable continental regions, *Seismol. Res. Lett.* **78**, no. 5, 554–562.
- Langer, C. J., M. G. Bonilla, and G. A. Bollinger (1987). Aftershocks and surface faulting associated with the intraplate Guinea, West Africa, earthquake of 22 December 1983, *Bull. Seismol. Soc. Am.* **77**, 1579–1601.
- Langston, C. A. (1976). A body wave inversion of Koyna, India earthquake of December 10, 1967 and some implications for body wave focal mechanisms, *J. Geophys. Res.* **81**, 2517–2529.
- Langston, C. A. (1981). Source inversion of seismic waveforms: The Koyna, earthquakes of 13 September 1967, *Bull. Seismol. Soc. Am.* **71**, 1–24.
- Leonard, M. (2008). One hundred years of earthquake recording in Australia, *Bull. Seismol. Soc. Am.* **98**, 1458–1470.
- Leonard, M. (2010). Earthquake fault scaling: Relating rupture length, width, average displacement, and moment release, *Bull. Seismol. Soc. Am.* **100**, no. 5, 1971–1988.
- Leonard, M. (2012). Erratum: Earthquake fault scaling: relating rupture length, width, average displacement, and moment release, *Bull. Seismol. Soc. Am.* **102**, 2797.
- Leonard, M., and D. Clark (2011). A record of stable continental region earthquakes from Western Australia spanning the late Pleistocene: Insights for contemporary seismicity, *Earth Planet. Sci. Lett.* **309**, 207–212.
- Leonard, M., D. Burbidge, T. Allen, D. Robinson, D. Clark, A. McPherson, and C. Collins (2014). The challenges of probabilistic seismic hazard assessment in stable continental interiors—An Australian example, *Bull. Seismol. Soc. Am.* **104**, no. 6, doi: [10.1785/0120130248](https://doi.org/10.1785/0120130248).
- Machette, M. N., A. J. Crone, and J. R. Bowman (1993). Geologic investigations of the 1986 Marryat Creek, Australia, earthquake—Implications for paleoseismicity in stable continental regions, *U.S. Geol. Surv. Bull.* **2032-B**, 29 pp.
- Madariaga, R. (1976). Dynamics of an expanding circular fault, *Bull. Seismol. Soc. Am.* **66**, no. 3, 639–666.
- Mandal, P., and S. Horton (2007). Relocation of aftershocks, focal mechanisms and stress inversion: Implications towards the seismo-tectonics of the causative fault zones of M_w 7.6 2001 Bhuj earthquake (India), *Tectonophysics* **429**, 61–78.
- Mandal, P., B. K. Rastogi, and C. S. P. Sarma (1998). Source parameters of Koyna earthquakes, India, *Bull. Seismol. Soc. Am.* **88**, no. 3, 833–842.
- Mandal, P., B. K. Rastogi, H. V. S. Satyanarayana, M. Kousalya, R. Vijayraghavan, C. Satyamurthy, I. P. Raju, A. N. S. Sarma, and N. Kumar (2004). Characterization of the causative fault system for the 2001 Bhuj earthquake of M_w 7.7, *Tectonophysics* **378**, 105–121.
- Manighetti, I., M. Campillo, S. Bouley, and F. Cotton (2007). Earthquake scaling, fault segmentation, and structural maturity, *Earth Planet. Sci. Lett.* **253**, nos. 3/4, 429–438.
- Mazzotti, S., and J. Adams (2005). Rates and uncertainties on seismic moment and deformation in eastern Canada, *J. Geophys. Res.* **110**, no. B09301, doi: [10.1029/2004JB003510](https://doi.org/10.1029/2004JB003510).
- McCalpin, J. P., and M. G. Thakkar (2003). 2001 Bhuj-Kachchh earthquake: Surface faulting and its relation with neotectonics and regional structures, Gujarat, Western India, *Ann. Geophys.* **46**, no. 5, 937–956.
- McCue, K. (1990). Australia's large earthquakes and recent fault scarps, *J. Struct. Geol.* **12**, 761–766.
- McNamara, D. E., H. M. Benz, R. B. Herrmann, E. A. Bergman, P. Earle, A. Meltzer, M. Withers, and M. Chapman (2014). The M_w 5.8 mineral, Virginia, earthquake of August 2011 and aftershock sequence: Constraints on earthquake source parameters and fault geometry, *Bull. Seismol. Soc. Am.* **104**, no. 1, 40–54.
- Nabelek, J., and G. Suarez (1989). The 1983 Goodnow earthquake in the central Adirondacks, New York: Rupture of a simple, circular crack, *Bull. Seismol. Soc. Am.* **79**, no. 6, 1762–1777.
- Nuttli, O. W. (1983). Average seismic source-parameter relations for mid-plate earthquakes, *Bull. Seismol. Soc. Am.* **73**, no. 2, 519–535.
- Pegler, G., and S. Das (1996). Analysis of the relationship between seismic moment and fault length for large crustal strike-slip earthquakes between 1977–1992, *Geophys. Res. Lett.* **23**, 905–908.
- Purcaru, G., and H. Berckhemer (1978). A magnitude scale for very large earthquakes, *Tectonophysics* **49**, 207–212.
- Quittmeyer, R. C., and K. H. Jacob (1979). Historical and modern seismicity of Pakistan, Afghanistan, northwestern India, and southeastern Iran, *Bull. Seismol. Soc. Am.* **69**, 773–823.
- Qureshi, I. R., and A. A. Sadig (1967). Earthquakes and associated faulting in central Sudan, *Nature* **215**, 263–265.
- Rajendran, K., C. P. Rajendran, M. Thakkar, and M. P. Tuttle (2001). The 2001 Kutch (Bhuj) earthquake: Coseismic surface features and their significance, *Curr. Sci.* **80**, 1397–1405.
- Rastogi, B. K., and P. Talwani (1980). Relocation of Koyna earthquakes, *Bull. Seismol. Soc. Am.* **70**, 1849–1868.
- Rockwell, T., J. Fonseca, C. Madden, T. Dawson, L. A. Owen, S. Vilanova, and P. Figueiredo (2009). Palaeoseismology of the Vilarica Segment of the Manteigas-Braganca fault in northeastern Portugal, *Geological Society, London, Special Publications* **316**, no. 1, 237–258.
- Romanowicz, B., and L. J. Ruff (2002). On moment-length scaling of large strike slip earthquakes and the strength of faults, *Geophys. Res. Lett.* **29**, no. 12, doi: [10.1029/2001GL014479](https://doi.org/10.1029/2001GL014479).
- Sato, R. (1972). Stress drop for a finite fault, *J. Phys. Earth* **20**, no. 4, 397–407.
- Scholz, C. H., C. A. Aviles, and S. G. Wesnousky (1986). Scaling differences between large interplate and intraplate earthquakes, *Bull. Seismol. Soc. Am.* **76**, 65–70.
- Schulte, S. M., and W. D. Mooney (2005). An updated global earthquake catalogue for stable continental regions: Reassessing the correlation with ancient rifts, *Geophys. J. Int.* **161**, 707–721.
- Seeber, L., J. G. Armbruster, and W. Y. Kim (2004). A fluid-injection-triggered earthquake sequence in Ashtabula, Ohio: Implications for seismogenesis in stable continental regions, *Bull. Seismol. Soc. Am.* **94**, 76–87.
- Seeber, L., G. Ekström, S. K. Jain, C. V. R. Murty, N. Chandak, and J. G. Armbruster (1996). The 1993 Killari earthquake in central India: A new fault in Mesozoic basalt flows? *J. Geophys. Res.* **101**, no. B4, 8543–8560.
- Seeber, L., W. Y. Kim, J. G. Armbruster, W. X. Du, and A. Lerner-Lam (2002). The 20 April 2002 M_w 5.0 earthquake near Au Sable Forks,

- Adirondacks, New York: A first glance at a new sequence, *Seismol. Res. Lett.* **72**, no. 4, 480–489.
- Shao, G., J. Crempien, R. J. Archeleta, and C. Ji (2011). Rupture process and stress drop of the 2011 M_w 5.8 Virginia earthquake based on seismic waveforms, *American Geophysical Union, Fall Meeting 2011*, San Francisco, California, 5–9 December 2011, Abstract S11B–2237.
- Shashidhar, D., N. P. Rao, D. Srinagesh, H. Gupta, H. V. S. Satyanarayana, G. Suresh, and A. Satish (2013). The 14 April 2012 Koyna earthquake of M_w 4.8: Insights into active tectonics of the Koyna region, *J. Seismol.* **17**, no. 4, 1345–1353.
- Shaw, B. E., and C. H. Scholz (2001). Slip-length scaling in large earthquakes; observations and theory and implications for earthquake physics, *Geophys. Res. Lett.* **28**, no. 15, 2995–2998.
- Shaw, B. E., and S. G. Wesnousky (2008). Slip-length scaling in large earthquakes: The role of deep-penetrating slip below the seismogenic layer, *Bull. Seismol. Soc. Am.* **98**, no. 4, 1633–1641.
- Singh, D. D., B. K. Rastogi, and H. K. Gupta (1975). Surface wave radiation pattern and source parameters of Koyna earthquake of December 10, 1967, *Bull. Seismol. Soc. Am.* **65**, 711–731.
- Somerville, P., K. Irikura, R. Graves, S. Sawada, D. Wald, N. Abrahamson, and A. Kowada (1999). Characterizing crustal earthquake slip models for the prediction of strong ground motion, *Seismol. Res. Lett.* **70**, no. 1, 59–80.
- Stirling, M., T. Goded, K. Berryman, and N. Litchfield (2013). Selection of earthquake scaling relationships for seismic-hazard analysis, *Bull. Seismol. Soc. Am.* **103**, no. 6, 2993–3011.
- Storchak, D. A., D. Di Giacomo, I. Bondár, E. R. Engdahl, J. Harris, W. H. K. Lee, A. Villaseñor, and P. Bormann (2010). Public release of the ISC-GEM Global Instrumental Earthquake catalogue (1900–2009), *Seismol. Res. Lett.* **84**, no. 5, 810–815, doi: [10.1785/0220130034](https://doi.org/10.1785/0220130034).
- Strasser, F. O., M. C. Arango, and J. J. Bommer (2010). Scaling of the source dimensions of interface and intraslab subduction-zone earthquakes with moment magnitude, *Seismol. Res. Lett.* **81**, no. 6, 940–950.
- Vogfjörð, K. S., and C. A. Langston (1987). The Meckering earthquake of 14th October 1968: A possible downward propagating rupture, *Bull. Seismol. Soc. Am.* **77**, 1558–1578.
- Wells, D. L., and K. J. Coppersmith (1994). New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement, *Bull. Seismol. Soc. Am.* **84**, no. 4, 974–1002.
- Wesnousky, S. G., L. Seeber, T. K. Rockwell, V. Thakur, R. Briggs, S. Kumar, and D. Ragona (2001). Eight days in Bhuj: Field report bearing on surface rupture and genesis of the 26 January 2001 earthquake in India, *Seismol. Res. Lett.* **72**, no. 5, 514–524.
- Wessel, P., and W. H. F. Smith (1998). New, improved version of the Generic Mapping Tools released, *Eos Trans. AGU* **79**, 579.
- Wheeler, R. L. (2009). Sizes of the largest possible earthquakes in the central and eastern United States—Summary of a workshop, *U.S. Geol. Surv. Open-File Rept. 2009-1263*, Golden, Colorado, 8–9 September 2008, 308 pp.
- Wheeler, R. L. (2014). Earthquake catalog for estimation of maximum earthquake magnitude, Central and Eastern United States—Part A, Prehistoric earthquakes, *U.S. Geol. Surv. Open-File Rept. 2014-1025-A*, 26 pp., doi: [10.3133/ofr20141025A](https://doi.org/10.3133/ofr20141025A).
- Yang, H., L. Zhu, and R. Chu (2009). Fault-plane determination of the 18 April 2008 Mount Carmel, Illinois, earthquake by detecting and relocating aftershocks, *Bull. Seismol. Soc. Am.* **99**, no. 6, 3413–3420.
- Yarwood, D. R., and D. I. Doser (1990). Deflection of oceanic transform motion at a continental margin as deduced from waveform inversion of the 1939 Accra, Ghana earthquake, *Tectonophysics* **172**, 341–349.
- (2007). The bulk of the rupture dimensions in these datasets were estimated from aftershock mapping, with the displacements and surface rupture lengths estimated from scarp data. Pegler and Das (1996) and Henry and Das (2001) both investigated the validity of using aftershocks to estimate rupture dimensions, and both concluded that aftershocks were a valid method for estimating rupture dimensions. Other methods include spectra fitting (Haskell, 1966; Brune, 1970; Sato, 1972; Madariaga, 1976) and waveform fitting (e.g., Somerville *et al.*, 1999). Somerville *et al.* (1999) suggested that waveform fitting tended to overestimate the rupture dimension, with a ring of low-slip tiles surrounding the fault rupture, and suggested cropping the outer tiles, thereby reducing the fault dimensions by 10%–20%. Three new stable continental region (SCR) earthquakes (2001 M_w 7.6, 2002 M_w 5.0, and 2011 M_w 5.74; discussed below) have used multiple methods to estimate the fault rupture dimensions. These suggest Brune spectra fitting overestimates stress drop and so underestimates the fault dimensions with aftershock analysis and waveform inversion, giving similar results. In compiling this database, I have given preference to aftershock analysis and rupture inversion. Where these are not available, I have used source duration and Brune spectra fitting. Where only Brune spectra fitting is available, I have taken this as minimum source dimension.
- Where available, I have used the M_w , using $M_w = (\log(M_0) - 9.1) / 1.5$ (Kanamori and Anderson, 1975; Kanamori, 1978; Purcaru and Berckhemer, 1978), from the International Seismological Centre-Global Earthquake Model (ISC-GEM; Storchak *et al.*, 2013) Catalogue.

New Reverse Faults in Stable Continental Regions

Goodnow, New York, M_b 5.1, M_w 4.8, October 1983

Nabelek and Suarez (1989) determined an M_0 of 1.9×10^{16} N·m (M_w 4.8) and, from the source duration, a radius of 0.7 ± 0.2 km with a stress drop of 26.5 (11.5–67) MPa. This source dimension is slightly smaller than the aftershock inferred dimension of 2.0 (1.5–3.0) and 1.8 (1.3–2.1) km. I choose 1.8×1.8 km² for the source dimensions.

Killari, Latur, India, M_w 6.17, 29 September 1993

From the analysis of aftershocks, Seeber *et al.* (1996), Harsh *et al.* (1998), Gupta *et al.* (1999), and Gahalaut *et al.* (2003) obtained similar results, with a fault length of 8 (6–10) km and width of 8 (6–9) km. Seeber *et al.* (1996) mapped a 1 km long zone of scarps with a second surface feature 3 km away. From leveling, Gupta *et al.* (1999) mapped a similar surface expression.

Ellalong, Australia, m_b 5.1, M_w 4.71, 6 August 1994

Aftershocks indicate a poorly constrained fault length of 4.0 (2.5–6.0) km (Jones *et al.*, 1994).

Appendix

Leonard (2010) relied on published datasets from Johnston (1994), Wells and Coppersmith (1994), Henry and Das (2001), Shaw and Scholz (2001), Hanks and Bakun (2002, 2008), Romanowicz and Ruff (2002), and Manighetti *et al.*

Eugowra, Australia, M_L 4.1, M_w 3.9, 21 August 1994

Because the 21 August event was preceded by numerous foreshocks, a network of eight temporary stations were in place to record the M_L 4.1 earthquake. Analysis of the foreshocks and aftershocks within eight days and 8–60 days (Gibson *et al.*, 1994) gives a 0.8 ± 0.2 km long $\times$ 0.9 ± 0.2 km wide fault with a dip of 38° .

Bhuj, India, M_w 7.63, 26 January 2001

From the analysis of aftershocks, Rajendran *et al.* (2001), Mandal *et al.* (2004), and Mandal and Horton (2007) obtained similar results, with a fault length of 32–60 km, width of 35–50 km, and dip of 45° . From combined inversion of teleseismic waveforms and geodetic data, Copley *et al.* (2011) estimated fault dimensions of 65 km $\times$ 35 km, with a 25×25 km² area of high slip. Although no major fault scarp was formed (Wesnousky *et al.*, 2001), Copley *et al.* (2011) estimated that slip at the surface along a length of 26 km to be 0.6–1.5 m, based on Interferometric Synthetic Aperture Radar (InSAR) and leveling data. The extensive areas of liquefaction limited the area of usable InSAR data (Copley *et al.*, 2011) and limited the areas where a scarp could have been mapped by field parties (e.g., McCaig and Thakkar, 2003). I choose 55 (40–65) km $\times$ 35 (30–50) km for the fault dimensions and do not estimate a surface rupture length.

Au Sable Forks, New York, M_w 4.95, 20 April 2002

This earthquake appears to be well constrained from the aftershock distribution; a fault-plane length of 2.8 (2.4–3.3) km and width of 2.4 (2.0–3.0) km was defined by Seeber *et al.* (2002). This is consistent with waveform modeling by Seeber *et al.* (2002) that indicated a circular rupture with a radius of 1.3 km, with the similarity of aftershock waveform indicating a simple flat fault plane. Abercrombie and Baise (2005) refined the aftershock relocations and obtained very similar dimensions, though their source spectra gave a radius of 0.8 km and a stress drop of 30 MPa.

Katanning, Western Australia, M_w 4.71, 10 October 2007

Modeling of satellite InSAR data (Dawson *et al.*, 2008) suggests a fault-plane length of 1.26 ± 0.05 km and width of 0.86 ± 0.08 km, with a stress drop of 30 ± 6 MPa. In the absence of supporting aftershock, waveform modeling, or even source spectra analysis, I consider these to be lower bound estimates and increase the upper uncertainty with a preferred dimensions of 1.26 (1.2–1.8) km and 0.9 (0.8–1.5). Vic Dent (personal comm., 2007) mapped a 0.25 km long fault surface trace that consisted of surface cracking and areas of localized uplift of ~ 0.3 m.

Mineral, Virginia, M_w 5.74, 23 August 2011

This earthquake occurred in a region of dense, by SCR standards, seismic networks, and a dense aftershock network was rapidly deployed. The open access to this data resulted in this earthquake sequence being studied using a range of techniques by several researchers. The aftershock distribution (Chapman, 2013; McNamara *et al.*, 2014) suggests a fault-plane length of 7 (6–9) km and width of 5.5 (4.5–7.0) km. From a finite-fault inversion, Shao *et al.* (2011) found 6 (5–7) km $\times$ 5 (4.5–5.5) km, with an average stress drop of 9 MPa, but with most of the moment release limited to less than 50% of the area. Waveform modeling (Chapman, 2013) suggests 70% of the moment was released in three subevents separated by 2–3 km. From source spectral fitting, Ellsworth *et al.* (2011) estimated a circular (i.e., Brune, 1970) source radius of 1.6–2.0 km with an average slip of 1.5–2.5 m and an average stress drop of 50–75 MPa. Using a similar technique, Cramer *et al.* (2011) determined a stress drop of 30 MPa. These results suggest the small source dimensions and high stress drop commonly reported for SCR earthquakes, from fitting Brune (1970) spectra, might be underestimating the source area and should be considered a minimum size. M_0 estimates are $3.8\text{--}6.0 \times 10^{17}$ N·m (M_w 5.65–5.79). Our preferred magnitude is 5.74. Our preferred dimensions are $L_{FR} = 6.5$ (5.5–8) and $W = 5$ (4.5–6.5).

Ernabella, South Australia, M_w 5.33, 23 March 2012

Clark *et al.* (2014) mapped a 1.6 km long surface-deformation zone. From source spectral analysis, Trevor Allen (unpublished, 2013) estimated a source radius of 1.93 (1.83–2.13) km. Given the results from the 2001 Virginia M_w 5.7 and 2002 Au Sable Forks M_w 5.0 earthquakes, I consider 3.4 km ($L = \sqrt{A} = \sqrt{(\pi r^2)}$) a minimum fault length and width. I assign 5.1 (3.4–6.8) as the fault rupture length.

Previously Published Stable Continental Region Faults

For several of the other well-documented SCR earthquakes (e.g., Johnston, 1994; Crone *et al.*, 2003; Schulte and Mooney, 2005), the surface rupture length (L_{SR}) is well defined, whereas the fault rupture length (L_{FR}) is not well constrained. Here, I summarize our estimation of the subsurface fault rupture length.

Allah Bund (Kutch), India, M_w 7.8, 16 June 1819

This fault has a surface rupture of 140 km (Quittmeyer and Jacob, 1979), and I assume for a fault of this size, the L_{FR} is equal to the L_{SR} .

Meckering, Australia, M_w 6.5, 14 October 1968

The surface rupture was accurately mapped (Gordon and Lewis, 1980), but the fault has not been defined seismologically, either from aftershocks or waveform inversion (Denham *et al.*, 1980; Vogfjörð and Langston, 1987). I assign a lower bound of the L_{FR} as 28 km (the distance between the endpoints of the surface fault), the upper bound of the L_{FR} as 41 km, (the scarp length +10%), and the mean of 35 km as the preferred L_{FR} .

Calingiri, Australia, M_w 5.7, 10 March 1970

The magnitude of this earthquake is poorly constrained with ISC-GEM giving M_w 5.71 ± 0.75 m_b 5.5. The surface rupture was accurately mapped (Gordon and Lewis, 1980), but neither aftershocks nor waveform modeling has defined a fault rupture length (Denham *et al.*, 1980).

Cadoux, Australia, M_w 6.1, 2 June 1979

The aftershocks suggest the $L_{FR} = L_{SR} = 14$ –16 km (Denham *et al.*, 1987). I assign an L_{FR} of 16 ± 2 km and L_{SR} of 14 (14–16) km. The magnitude is M_w 6.1 ± 0.1 .

Guinea, Africa, M_w 6.3, 22 December 1983

Langer *et al.* (1987) measured a scarp length of 9.4 (9.0–9.6) km and from aftershocks an elliptical rupture area of 22 (20–26) km $\times$ 10 (8–12) km. Because the aftershock recording started 27 days after the mainshock, this should be considered an upper limit. Transforming this ellipse into a rectangle gives 19 (16–24) km $\times$ 9 (8–10) km. Dorbath *et al.* (1984) obtained similar results for both scarp length and aftershock dimension. Two of the four focal mechanisms (Langer *et al.*, 1987) are predominately normal with significant strike slip, and two are strike slip with significant normal. Because of this ambiguity, I plot this earthquake on the $M_0 \sim L$ plots for both dip slip and strike slip but do not use it in the regressions. The magnitude is M_w 6.29 ± 0.1 .

Marryat Creek, Australia, M_w 5.7, 30 March 1986

The surface rupture has a length of 11 (9.0–13.5) km (Machette *et al.*, 1993). McCue (1990) suggested the fault length is 9 km, the length between the endpoints of the surface rupture. I consider this a lower limit and choose 11 km as the preferred value, with the surface trace (13 km) being the upper limit. The magnitude is M_w 5.71 ± 0.1 .

Tennant Creek, Australia, M_w 6.2, 6.3, 6.6, 22 January 1988

Choy and Bowman (1990) estimated the seismic moment of the three earthquakes to be 2.46×10^{18} (M_w 6.19), 3.02×10^{18} (M_w 6.25), and $7.59 \pm 0.8 \times 10^{18}$ N·m (M_w 6.53), respectively. The ISC-GEM magnitudes are 0.05–0.05 magnitude units higher for all three earthquakes. The three causative faults have been accurately determined

from a combination of aftershock analysis (Bowman, 1988; Bowman *et al.*, 1990) and modeling of the elevation profiles (Bowman, 1991) at 13 (7–13), 10 (6–10), and 15 (12–19) km. The surface rupture lengths (Bowman, 1992; Crone *et al.*, 1992) are 10.2 (10–10.5), 6.7 (6.5–6.9), and 16.0 (15.5–16.5) km, respectively. The magnitudes are M_w 6.23 ± 0.1 , 6.3 ± 0.1 , and 6.58 ± 0.1 .

Ungava, Canada, M_w 6.0, 25 December 1989

Adams *et al.* (1991, 1992; cited in Bent, 1994) mapped the surface rupture, giving $L_{SR} = 9.5$ (9.0–10.5) km. From a combination of aftershock and waveform modeling, the fault is thought to extend another few kilometers beyond the mapped surface rupture (Bent, 1994). I assign 12.5 ± 1.5 km. The magnitude is M_w 5.97 ± 0.1 .

Strike-Slip Faults in Stable Continental Region

Johnston (1994) identified two SCR strike-slip earthquakes: the 1939 M_w 6.5 Ghana, Africa, and 1966 M_w 5.6 Sudan, Africa, events. Several other SCR strike-slip earthquakes have since been identified.

Ghana, Africa, M_w 6.5, 22 July 1939

The surface scarp is 13 (13–20) km (Bates, 1941; Yarwood and Doser, 1990). The magnitude is M_w 6.45.

Sudan, Africa, M_w 5.6, 9 October 1966

The surface scarp is 4 (4–7) km (Qureshi and Sadig, 1967). The magnitude is M_w 5.64.

Koyna, India, M_w 6.3, 10 December 1967

Using body-wave inversion, Langston (1976, 1981) proposed an almost pure strike-slip event with an M_0 of $3.2 \pm 1.4 \times 10^{18}$ N·m. One of the focal plains corresponds very closely with the >1300 earthquakes located in the region by a dense local network (Mandal *et al.*, 1998). The total length of this seismicity is 28 km, though Mandal *et al.* (1998) suggests only the 15 km long northern cluster should be considered aftershocks of the 1967 earthquake. This is consistent with the findings of Gupta *et al.* (1980) and Rastogi and Talwani (1980). From surface-wave modeling, Singh *et al.* (1975) estimated a fault length of 25 km, with a range of 23–40 km for rupture velocities of 1.5 and 3.0 km/s, respectively. I choose a length of 18 (15–25) km and a width of 8 (5–12) km from the seismicity profiles of Mandal *et al.* (1998). The earthquake caused a 3 ± 0.5 km long surface feature (Seeber *et al.*, 1996; Gupta *et al.*, 1999) that is coincident with the northern cluster.

Ashtabula, Ohio, M_w 3.8, 31 January 1987 and M_w 4.3, 26 January 2001

These two earthquakes are considered to be induced by fluid injection (Seeber *et al.*, 2004) and so are not included in this analysis.

Mt. Carmel, Illinois, M_w 5.2, 18 April 2008

Several estimates of the M_w have been made: Herrmann *et al.* (2008) of 5.23 (M_0 8.8×10^{16} N·m), Global Centroid Moment Tensor (Global CMT) of 5.33 (M_0 12.4×10^{16} N·m), and U.S. Geological Survey (USGS) of 5.19 (M_0 7.6×10^{16} N·m). From a precise relocation of the aftershocks, Yang *et al.* (2009) estimated a fault length of 5.0 (3.5–6.0) km and width of 6.2 (5.0–7.0) km. Finite-fault models by Hartzell and Mendoza (2011) suggest a $3 \text{ km} \times 2 \text{ km}$ asperity with a poorly constrained ($\sim 6 \times 5 \text{ km}$) area and estimates of moment ranging from 7.0 to 17×10^{16} N·m, with a preferred value of 8.8×10^{16} N·m (M_w 5.23). I assign M_w 5.23 $8.8 (7.0\text{--}17) \times 10^{16}$ N·m, L_{FR} 5 (3–6) km, and W 6.0 (5–7) km.

Siheung, Korea, M_w 3.14, 9 February 2010

From the rise time, Kim and Kim (2014) estimated a fault radius of 0.12 ± 0.01 km and, from relocated aftershocks, an L_{FR} of 0.24 (0.2–0.3) km and W of 0.15 (0.1–0.2) km. Their estimate for moment was $6.5 \pm 0.35 \times 10^{13}$ N·m. They also estimated the radius of a foreshock on 7 August 2004 as 0.11 ± 0.01 km with an M_0 of 1.06×10^{13} N·m (M_w 2.62).

Prague, Oklahoma, M_w 5.7, 6 November 2011

Three earthquakes with M_w of 5.0, 5.7, and 5.0 occurred in Oklahoma over a three-day period in November 2011. As seismometers were deployed within 24 hours of the first earthquake on 5 November, the second and third $M_w \geq 5.0$ earthquakes, as well as thousands of aftershocks, were accurately recorded. This enabled Keranen *et al.* (2013) to illuminate the faults from the temporal and spatial distribution of aftershocks. The three faults are estimated as: L_{FR} 4.0 (3.5–5.0) and W 4.5 (3–5), L_{FR} 6.0 (5–8) and W 5.0 (4–7), and L_{FR} 3.2 (3.0–4.0) and W 4.0 (2–6) km.

Koyno, India, M_w 4.8, 14 April 2012

From the location of aftershocks, Shashidhar *et al.* (2013) suggest a fault plane of $10 \text{ km} \times 10 \text{ km}$, with the aftershocks, first motion, and P -wave inversion all consistent with a steeply dipping north–south-trending fault. Shashidhar *et al.* (2013) suggest this is on the same fault plane as the 1967 M_w 6.3 earthquake. As its aftershocks appear to completely overlap with the aftershocks of the 1967 earthquake, I consider it an aftershock and do not include it in the analysis.

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